CONCEPT 47.1

Fertilization and cleavage initiate embryonic development

Development occurs at multiple points in the animal life cycle. In a frog, for example, the larva (tadpole) undergoes sweeping changes in anatomy in becoming an adult. Development occurs in adult animals too, as when stem cells in the gonads produce sperm and eggs (gametes). In this chapter, however, our focus is on development in the embryonic stage.

Embryonic development in many animal species involves common stages in a set order. The first is fertilization, the fusion of sperm and egg. Next is cleavage, a series of cell divisions that divide, or cleave, the embryo into many cells. These cleavage divisions, which typically are rapid and lack accompanying cell growth, generate a hollow ball of cells called a blastula. In gastrulation, this blastula folds in on itself, rearranging into a multilayered embryo, the gastrula. Then, during organogenesis, local changes in cell shape and large-scale changes in cell location generate the rudimentary organs.

Because embryonic development has many common features across the animal kingdom, lessons learned from studying one animal can often be applied quite broadly. For this reason, studying development lends itself well to the use of **model organisms**, species chosen for the ease with which they can be studied. For example, the sea urchin (phylum *Echinodermata*) is useful for studying cell-surface events that take place during fertilization. Sea urchin gametes are easy to collect and fertilization occurs outside the body. As a result, researchers can observe fertilization and development in the laboratory simply by combining eggs and sperm in seawater.

In this chapter, we will concentrate on the sea urchin and four other model organisms: the fruit fly, the frog, the chick, and the nematode (roundworm). We will also explore some aspects of human embryonic development. We'll begin with the events surrounding **fertilization**, the formation of a diploid zygote from a haploid egg and sperm.

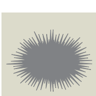
Fertilization

When sea urchins release their gametes into the water, the jelly coat of the egg exudes soluble molecules that attract the sperm, which swim toward the egg.

The Acrosomal Reaction

As detailed in **Figure 47.2**, the moment a sperm head contacts the egg surface, molecules in the jelly coat trigger

▼ **Figure 47.2 The acrosomal and cortical reactions during sea urchin fertilization.**



This icon is a simplified drawing of an adult sea urchin. Throughout the chapter, this icon and others of an adult frog, chicken, nematode, and human indicate the source of embryos in figures.